

FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration

TAHA SOLIMAN, [Robert Bosch GmbH](#)

NELLIE LALENI, [Fraunhofer IPMS](#), Center Nanoelectronic Technologies

TOBIAS KIRCHNER, [Robert Bosch GmbH](#)

FRANZ MÜLLER, ASHISH SHRIVASTAVA, and THOMAS KÄMPFE, [Fraunhofer IPMS](#), Center Nanoelectronic Technologies

ANDRE GUNTORO, [Robert Bosch GmbH](#)

NORBERT WEHN, [University of Kaiserslautern](#)

Today, a large number of applications depend on deep neural networks (DNN) to process data and perform complicated tasks at restricted power and latency specifications. Therefore, processing-in-memory (PIM) platforms are actively explored as a promising approach to improve the throughput and the energy efficiency of DNN computing systems. Several PIM architectures adopt resistive non-volatile memories as their main unit to build crossbar-based accelerators for DNN inference. However, these structures suffer from several drawbacks such as reliability, low accuracy, large ADCs/DACs power consumption and area, high write energy, and so on. In this article, we present a new mixed-signal in-memory architecture based on the bit-decomposition of the multiply and accumulate (MAC) operations. Our in-memory inference architecture uses a single FeFET as a non-volatile memory cell. Compared to the prior work, this system architecture provides a high level of parallelism while using only 3-bit ADCs. Also, it eliminates the need for any DAC. In addition, we provide flexibility and a very high utilization efficiency even for varying tasks and loads. Simulations demonstrate that we outperform state-of-the-art efficiencies with 36.5 TOPS/W and can pack 2.05 TOPS with 8-bit activation and 4-bit weight precision in an area of 4.9 mm² using 22 nm FDSOI technology. Employing binary operation, we obtain 1169 TOPS/W and over 261 TOPS/W/mm² on system level.

CCS Concepts: • **Hardware** → **Analog and mixed-signal circuits**; • **Computer systems organization** → **Neural networks**;

Additional Key Words and Phrases: FeFET crossbar array, in-memory computation, convolution neural networks, mixed-signal processing, bit-decomposition, current-mode ADC

This work received funding within the ECSEL Joint Undertaking project TEMPO in collaboration with the European Union's H2020 Framework Program (H2020/2014-2020) and National Authorities, under grant agreement number 826655 and was partially funded by the Carl Zeiss Foundation under the grant Sustainable Embedded AI.

Authors' addresses: T. Soliman, T. Kirchner, and A. Guntoro, [Robert Bosch GmbH](#), Renningen 71272, Germany; emails: {taha.soliman, tobias.kirchner, andre.guntoro}@de.bosch.com; N. Laleni, F. Müller, A. Shrivastava, and T. Kämpfe, [Fraunhofer IPMS](#), Center Nanoelectronic Technologies, Dresden, Germany; emails: {nelli.laleni, franz.mueller, ashish.shrivastava, thomas.kaempfe}@ipms.fraunhofer.de; N. Wehn, [University of Kaiserslautern](#), Kaiserslautern 67663, Germany; email: wehn@eit.uni-kl.de.



This work is licensed under a [Creative Commons Attribution International 4.0 License](#).

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

1539-9087/2022/10-ART84

<https://doi.org/10.1145/3529760>

ACM Transactions on Embedded Computing Systems, Vol. 21, No. 6, Article 84. Publication date: October 2022.